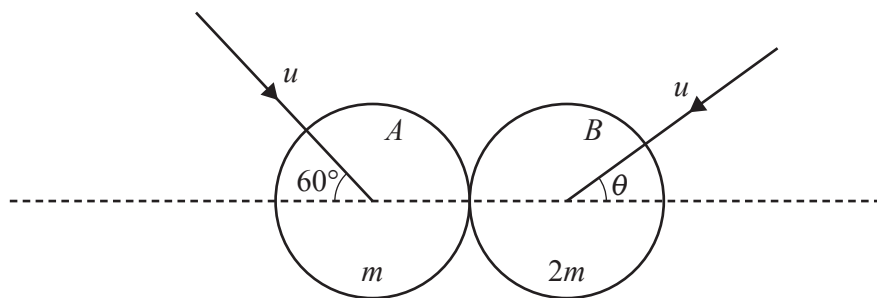


1



Two uniform smooth spheres A and B of equal radii have masses m and $2m$ respectively. The two spheres are moving with equal speeds u on a smooth horizontal surface when they collide. Immediately before the collision, A 's direction of motion makes an angle of 60° with the line of centres, and B 's direction of motion makes an angle θ with the line of centres (see diagram). The coefficient of restitution between the spheres is e .

After the collision, the component of the velocity of A along the line of centres is v and B moves perpendicular to the line of centres. Sphere A now has twice as much kinetic energy as sphere B .

- (a) Show that $v = \frac{1}{2}u(4\cos\theta - 1)$. [1]

By CoM: (horizontally)

A must change direction

$$mu\cos 60 - 2mu\cos\theta = m(-v) \Rightarrow \text{where rightward is positive}$$

$$v = 2u\cos\theta - \frac{u}{2}$$

$$v = \frac{1}{2}u(4\cos\theta - 1)$$

- (b) Find the value of $\cos\theta$. [4]

sphere A KE:

$$= \frac{1}{2}m \left(v^2 + (u\sin 60)^2 \right)$$

$$= \frac{1}{2}m \left(v^2 + u^2 \left(\frac{3}{4} \right) \right)$$

$$= \frac{1}{2}m \left(\left(\frac{u}{2}(4\cos\theta - 1) \right)^2 + \frac{3u^2}{4} \right)$$

$$= \frac{u^2}{4} (16\cos^2\theta - 8\cos\theta + 1 + 3)$$

KE_B:

$$\frac{1}{2}(2m)(u^2\sin^2\theta)$$

$$= mu^2\sin^2\theta$$

$$2 \cancel{u} u^2 \sin^2 \theta = \frac{1}{2} \cancel{u} \left(2u \cos \theta - \frac{u}{2} \right)^2 + \frac{3u^2}{4}$$

$$4 \cancel{u} \sin^2 \theta = 4 \cancel{u} \cos^2 \theta + \frac{u^2}{4} - 2 \cancel{u} \cos \theta + \frac{3u^2}{4}$$

$$4(1 - \cos^2 \theta) = 4 \cos^2 \theta - 2 \cos \theta + 1$$

$$0 = 8 \cos^2 \theta - 2 \cos \theta - 3$$

$$= (4 \cos \theta - 3)(2 \cos \theta + 1)$$

$$\rightarrow \cos \theta = \frac{3}{4} \text{ or } -\frac{1}{2}$$

We know $v_{\text{up}} > 0$

$$\frac{1}{2} (4u \cos \theta - 1) > 0$$

only when

$$\cos \theta = \frac{3}{4}$$

(c) Find the value of e .

[2]

$$e = \frac{v - 0}{u \cos 60 + u \cos \theta} = \frac{\frac{u}{2} (4(\frac{3}{4}) - 1)}{\frac{u}{2} + \frac{3u}{4}} = \frac{4}{5}$$

- 2 A ball of mass 2 kg is projected vertically downwards with speed 5 m s^{-1} through a liquid. At time t s after projection, the velocity of the ball is $v \text{ m s}^{-1}$ and its displacement from its starting point is x m. The forces acting on the ball are its weight and a resistive force of magnitude $0.2v^2 \text{ N}$.

(a) Find an expression for v in terms of t .

[6]

$$m \frac{dv}{dt} = mg - 0.2v^2$$

$$\frac{dv}{dt} = g - 0.1v^2$$

$$\frac{dv}{100 - v^2} = \frac{1}{10} dt$$

$$\int \frac{1}{(10-v)(10+v)} dv = \int \frac{1}{10} dt$$

$$\frac{1}{(10-v)(10+v)} = \frac{A}{(10-v)} + \frac{B}{(10+v)}$$

$$\hookrightarrow 10A + 10vA + 10B - 10vB$$

$$\hookrightarrow 10A - 10B = 0 \Rightarrow A = B$$

$$10A + 10B = 1 \Rightarrow A = B = \frac{1}{20}$$

$$\Rightarrow \frac{1}{20} \left(\int \frac{1}{10-v} dv + \int \frac{1}{10+v} dv \right) = \int \frac{1}{10} dt$$

$$\frac{\ln(10-v)}{(-1)} + \frac{\ln(10+v)}{(1)} = 2t + c$$

$$\ln\left(\frac{10+v}{10-v}\right) = 2t + c$$

$$\hookrightarrow \text{subst } (t, v) = (0, 5)$$

$$\ln 3 = c$$

$$\frac{10+v}{10-v} = 3e^{2t}$$

$$10+v = 30e^{2t} - 3ve^{2t}$$

$$v(1+3e^{2t}) = 30e^{2t} - 10$$

$$v = \frac{10(3e^{2t} - 1)}{3e^{2t} + 1}$$

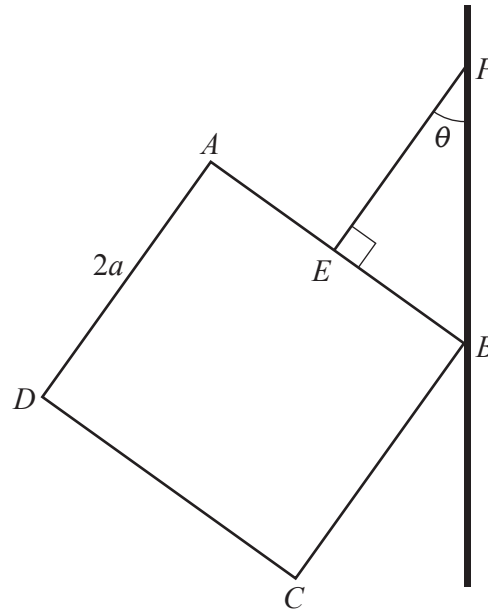
(b) Deduce what happens to v for large values of t .

[1]

$$v = \frac{10(3 - e^{-2t})}{3 + e^{-2t}}$$

$$\hookrightarrow \text{as } t \rightarrow \infty$$

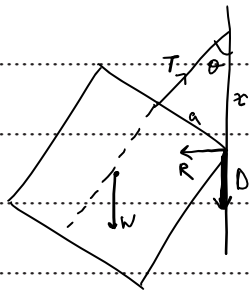
$$= \frac{10(3)}{3} = 10 \text{ ms}^{-1}$$



A uniform square lamina of side $2a$ and weight W is suspended from a light inextensible string attached to the midpoint E of the side AB . The other end of the string is attached to a fixed point P on a rough vertical wall. The vertex B of the lamina is in contact with the wall. The string EP is perpendicular to the side AB and makes an angle θ with the wall (see diagram). The string and the lamina are in a vertical plane perpendicular to the wall. The coefficient of friction between the wall and the lamina is $\frac{1}{2}$.

Given that the vertex B is about to slip up the wall, find the value of $\tan \theta$.

[8]



Translationally:

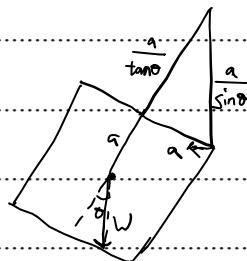
$$T \cos \theta = W + D \Rightarrow T \cos \theta = W + \frac{R}{2} \quad (1)$$

$$T \sin \theta = R \quad (2)$$

Rotationally:

$$D = \mu R$$

$$D = \frac{R}{2}$$



$$W \sin \theta \left(\frac{a}{\tan \theta} + a \right) = \frac{Ra}{\sin \theta} \quad (3)$$

Subst. ② in ③

With ① & ②:

$$W \sin \left(\frac{a}{\tan \theta} + d \right) = T_a$$

$$T \cos \theta - \frac{T \sin \theta}{2} = W$$

$$W \sin \left(\frac{\cos \theta}{\sin \theta} + 1 \right) = T$$

$$T \left(\cos \theta - \frac{\sin \theta}{2} \right) \left(\cos \theta + \sin \theta \right) = T$$

$$\cos^2 \theta + \frac{\sin \theta \cos \theta}{2} - \frac{\sin^2 \theta}{2} = 1$$

$$1 - \sin^2 \theta + \frac{\sin \theta \cos \theta}{2} - \frac{\sin^2 \theta}{2} = 1$$

$$\frac{-3\sin^2 \theta}{2} + \frac{\sin \theta \cos \theta}{2} = 0$$

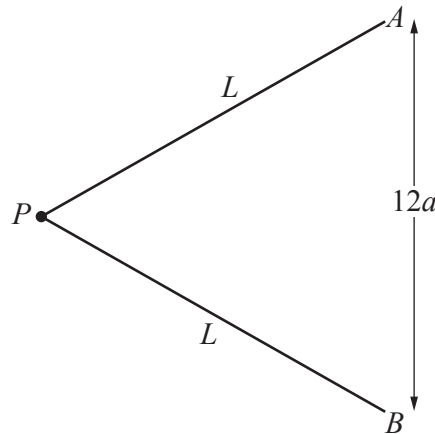
$$3\sin^2 \theta - \sin \theta \cos \theta = 0$$

$$\sin \theta (3\sin \theta - \cos \theta) = 0$$

$$\hookrightarrow \sin \theta = 0 \Rightarrow \text{evidently, impossible}$$

$$\hookrightarrow 3\sin \theta = \cos \theta$$

$$\hookrightarrow \boxed{\tan \theta = \frac{1}{3}}$$



A light elastic string has natural length $8a$ and modulus of elasticity $5mg$. A particle P of mass m is attached to the midpoint of the string. The ends of the string are attached to points A and B which are a distance $12a$ apart on a smooth horizontal table. The particle P is held on the table so that $AP = BP = L$ (see diagram). The particle P is released from rest. When P is at the midpoint of AB it has speed $\sqrt{80ag}$.

(a) Find L in terms of a .

[5]

Initially :

$$\begin{aligned} \text{EPE} &= \frac{\lambda x^2}{2l} \\ &= \frac{(5mg)(2L-8a)^2}{16a} \end{aligned}$$

$$\text{KE} = 0$$

Finally

$$\begin{aligned} \text{EPE} &= \frac{\lambda x^2}{2l} \\ &= \frac{5mg(4a)^2}{2(8a)} \\ &= 5mg \cdot a \end{aligned}$$

$$\text{KE} = \frac{1}{2} m (\sqrt{80a})^2 = 40mga$$

By CoE:

$$\left(\frac{5mg}{16a} \right) (2L-8a)^2 = 40mga + 5mga$$

$$(2L-8a)^2 = 16a(45a) = 720a^2$$

$$2L-8a = \pm 12a \Rightarrow \text{not } = -12a$$

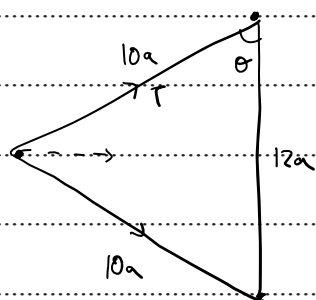
$\hookrightarrow$ then $L < 0$

$$\hookrightarrow 2L-8a = 12a$$

$$\boxed{L = 10a}$$

(b) Find the initial acceleration of P in terms of g .

[3]



$$F = 2T \sin \theta$$

$$\hookrightarrow \sin \theta = 0.8$$

$$\hookrightarrow T = \frac{\lambda x}{l} = \frac{(5mg)(12a)}{2 \times 8a}$$

$$= 7.5mg$$

$$F = 2(7.5mg)(0.8)$$

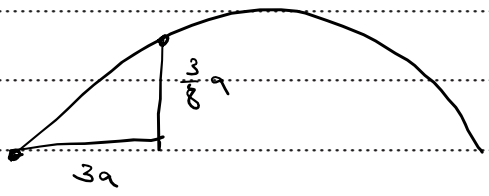
$$= 12mg$$

$$\hookrightarrow \boxed{a = 12g}$$

- 5 A particle P is projected with speed $u \text{ ms}^{-1}$ at an angle θ above the horizontal from a point O on a horizontal plane and moves freely under gravity. During its flight P passes through the point which is a horizontal distance $3a$ from O and a vertical distance $\frac{3}{8}a$ above the horizontal plane. It is given that $\tan \theta = \frac{1}{3}$.

(a) Show that $u^2 = 8ga$.

[2]



Handwritten solution for part (a):

$$y = x \tan \theta - \frac{gx^2}{2u^2} \sec^2 \theta$$

$$\hookrightarrow \text{substituting } (x, y, \tan \theta) = (3a, \frac{3}{8}a, \frac{1}{3})$$

$$\frac{3}{8}a = 3a \left(\frac{1}{3} \right) - \frac{g(9a^2)}{2u^2} (\tan^2 \theta + 1)$$

$$\frac{5ga^2}{2u^2} = \frac{5}{8}a$$

$$5ga^2 = \frac{5}{8}u^2a$$

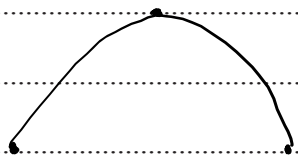
$$\hookrightarrow \boxed{u^2 = 8ga}$$

A particle Q is projected with speed $V \text{ ms}^{-1}$ at an angle α above the horizontal from O at the instant when P is at its highest point. Particles P and Q both land at the same point on the horizontal plane at the same time.

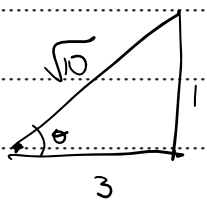
(b) Find V in terms of a and g .

[7]

for P :

$$\tan \theta = \frac{1}{3}, \quad u = \sqrt{8ga}$$


We know that horizontally, Q is twice as fast as P



$$\hookrightarrow V \cos \alpha = 2\sqrt{8ga} \cos(\tan^{-1}(\frac{1}{3}))$$

$$V \cos \alpha = \sqrt{288ga} \quad \text{①}$$

@ time = t , height of P = 0

$$\hookrightarrow s = vt + \frac{1}{2}at^2$$

$$0 = (\sqrt{8ga} \sin \theta)(t) - \frac{1}{2}g(t)^2$$

$$\sqrt{8ga} \left(\frac{1}{\sqrt{10}}\right) = 5t$$

$$t = \frac{\sqrt{0.8ga}}{5}$$

For Q:

$\hookrightarrow$ it reaches $s=0$ in half the time

$$0 = v\left(\frac{t}{2}\right)^2 + \frac{1}{2}a\left(\frac{t}{2}\right)^2$$

$$= v \sin d \left(\frac{\sqrt{0.8ga}}{10}\right) - \frac{1}{2}g\left(\frac{\sqrt{0.8ga}}{10}\right)^2$$

$$v \sin d = 5 \left(\frac{\sqrt{0.8ga}}{10}\right)$$

$$= \sqrt{0.2ga}$$

$$v = \sqrt{(v \sin d)^2 + (v \cos d)^2}$$

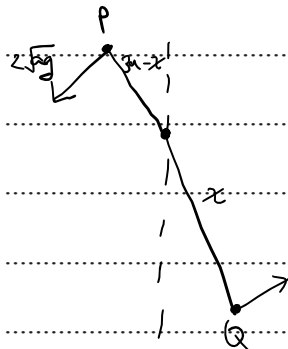
$$= \sqrt{28.8ga + 0.2ga}$$

$$= \sqrt{29ga}$$

- 6 A particle P of mass m is attached to one end of a light inextensible rod of length $3a$. An identical particle Q is attached to the other end of the rod. The rod is smoothly pivoted at a point O on the rod, where $OQ = x$. The system, of rod and particles, rotates about O in a vertical plane.

At an instant when the rod is vertical, with P above Q , the particle P is moving horizontally with speed u . When the rod has turned through an angle of 60° from the vertical, the speed of P is $2\sqrt{ag}$, and the tensions in the two parts of the rod, OP and OQ , have equal magnitudes.

- (a) Show that the speed of Q when the rod has turned through an angle of 60° from the vertical is $\frac{2x}{3a-x}\sqrt{ag}$. [2]



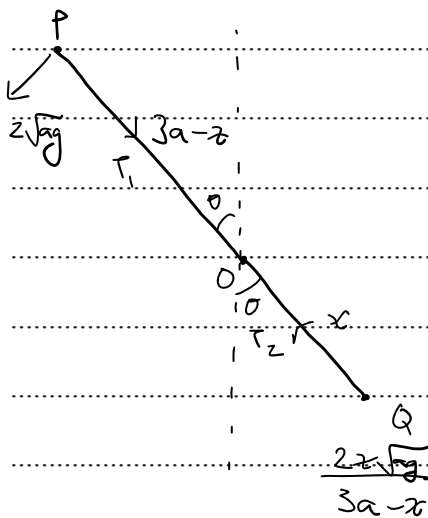
ω is constant for the system

$$\frac{v_P}{r_P} = \frac{v_Q}{r_Q}$$

$$\frac{2\sqrt{ag}}{3a-x} = \frac{v_Q}{x}$$

$$\hookrightarrow v_Q = \frac{2x\sqrt{ag}}{3a-x}$$

- (b) Find x in terms of a . [5]



@ P:

$$mg \cos \theta + T_1 = \frac{mv^2}{r}$$

$$T_1 = \frac{m(4ag)}{3a-x} - mg \cos 60$$

$$= \frac{4mga}{3a-x} - \frac{mg}{2}$$

@ Q:

$$\frac{mv^2}{r} = T_2 - mg \cos \theta$$

$$T_2 = \frac{m(4x^2 ag)}{(3a-x)^2 x} + \frac{mg}{2}$$

$$T_1 = T_2$$

$$\hookrightarrow \frac{4mga}{3a-x} - \frac{mg}{2} = \frac{4xmg}{(3a-x)^2} + \frac{mg}{2}$$

$$\frac{(3a-x)4a - 4xa}{(3a-x)^2} = 1$$

$$12a^2 - 4ax - 4xa = (3a-x)^2$$

$$12a^2 - 8ax = 9a^2 + x^2 - 6ax$$

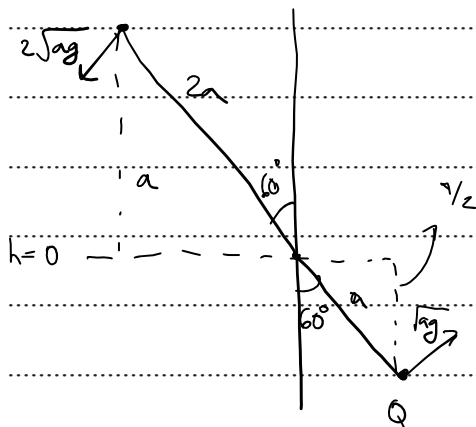
$$3a^2 - 2ax - x^2 = 0$$

$$(a-x)(3a+x) = 0$$

$$\hookrightarrow \boxed{x=a}$$

(c) Find u in terms of a and g .

[4]



At 60° :

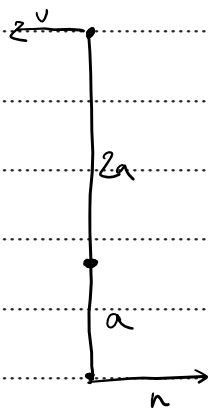
$$PE = mga + mg\left(-\frac{1}{2}\right) = \frac{mga}{2}$$

$$KE = \frac{1}{2}m(2\sqrt{5})^2 + \frac{1}{2}m(\sqrt{5})^2$$

$$= 2mga + \frac{1}{2}mga$$

$$= \frac{5}{2}mga$$

$$\hookrightarrow \text{Total energy} = 3mga$$



@ $\sigma = 0^\circ$

$$PE = -mga + 2mga$$

$$= mga$$

$$KE = 3mga - mga = 2mga$$

$$\rightarrow v = 2u$$

$$\hookrightarrow 2mga = \frac{1}{2}m(u^2 + (u)^2)$$

$$2mga = \frac{1}{2}m\left(\frac{5u^2}{4}\right)$$

$$u^2 = \frac{16}{5}a \Rightarrow \boxed{u = \frac{4}{\sqrt{5}}\sqrt{ga} \text{ ms}^{-1}}$$

[illegible]

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